# Photon and bit allocation in snapshot spectral sensing: the role of quantizer calibration

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**Abstract**

A sensor with a fixed output budget can record a few spectral channels precisely or more channels with fewer intensity levels. Photon-efficient routing makes the second option attractive, but the answer also depends on how photon counts are digitized and reconstructed. This study compares six channel–bit allocations at 24 bits per spatial macro-pixel using 15 scenes from the CAVE dataset. A Poisson forward model represents a spectral router and an equal-area absorptive filter over exposures from 10 to $10^5$ white-reference photons. Entire scenes are separated for training, validation and testing. The preferred allocation changes with the decoder and the quantizer, even when the optical model is fixed. At $N = 10^5$, validation-based joint selection of channel count and quantizer reduces router reconstruction RMSE by 24.0% relative to a fixed $3 \times 8$ uniform-ADC baseline, and by 13.1% relative to selecting channel count alone. A training-calibrated quantile control uses one-bit codes more evenly, but does not consistently improve spectral reconstruction. The results support selecting spectral sampling, intensity coding and reconstruction together; they do not establish a universal optimum based on spectral rank. The contribution is a reproducible comparison and design-selection procedure under explicit assumptions, rather than a new optical device.

**Keywords:** computational imaging; spectral routing; photon noise; quantization; sensor design

## 1 Introduction

A 24-bit RGB image stores three intensity values with eight bits each. The same output budget could instead hold six values with four bits each, or 24 values with one bit each. These representations have equal size, but they preserve different information. For spectral reconstruction, the useful question is not how many channels can be built. It is how many can be measured and coded accurately enough to improve the recovered spectrum.

Three effects compete. Broad channels average away spectral structure. Narrower channels receive fewer photons, increasing relative shot noise. Reducing the number of bits merges distinct intensities into the same output code. A reconstruction model can use correlations learned from other spectra, so the balance also depends on what the decoder knows. A channel-count recommendation that ignores the decoder or the ADC may therefore describe the implementation used to obtain it rather than a general property of the scene.

This trade-off is not new in its broad form. Laska and Baraniuk studied measurement count versus bit depth under a fixed bit budget in compressive sensing [1]. Özçelikkale and Ozaktas examined the number and precision of samples used to represent an optical field [2]. Neither result transfers directly to the present model: here, changing the number of spectral bands also changes the detected photon count per measurement. In hyperspectral imaging, photon efficiency, spectral structure and quantization already form part of the established design discussion [3].

Spectral acquisition has also been optimized at the camera level. The generalized assorted pixel camera of Yasuma et al. combines spectral and exposure diversity in a spatial mosaic [4]. More recently, van Elst et al. optimized integrated spectral sensors for specific applications and demonstrated compact devices with only a few sensing elements [5]. These studies motivate evaluating the acquisition design against the task it is meant to perform.

Routing introduces a useful additional variable. A wavelength router directs light toward different detector outputs, while an absorptive filter rejects out-of-band light. Zou et al. reported an RGB metasurface router with 84% average energy utilization [6]. Han et al. subsequently studied the optical scaling of multichannel routers with electromagnetic simulations [7]. Optical efficiency alone, however, does not specify how the surviving counts should be divided among finite digital codes.

The present study asks which channel–bit allocation minimizes reconstruction error on an unseen scene at fixed area, exposure and output budget. It then tests whether that recommendation survives changes in the decoder and ADC. The study combines an explicit forward model with scene-held-out design selection and controls that separate photon throughput from coding. A targeted quantizer experiment examines whether poor low-bit performance reflects unused code range. All counts are simulated from measured scene spectra.

# 2 Forward model and budget

## 2.1 What is held fixed

Let $x \in [0,1]^{31}$ denote a normalized reference spectrum at one spatial location. A macro-pixel contains $K$ detector outputs, each assigned $B$ bits. The primary constraint is

$$KB = 24, \qquad (K,B) \in \mathcal{D}, \tag{1}$$

where $\mathcal{D} = \{(3,8),(4,6),(6,4),(8,3),(12,2),(24,1)\}$. The three-channel candidate uses three contiguous spectral bands; it is a reference design, not a model of a commercial RGB camera and its processing pipeline.

A nonnegative matrix $P_K \in \mathbb{R}^{K\times 31}$ partitions the source bands into contiguous groups. Each source band contributes a total weight of $1/31$, so the column sums of $P_K$ are $1/31$. The primary construction assigns each source band to one output. Group sizes differ by at most one source band. A separate equal-width overlap construction tests this discretization choice.

The exposure parameter $N$ is the expected incident photon count for a unit-valued white reference before optical losses. Actual scene pixels have different expected counts according to $x$. This fixes exposure across designs without artificially assigning the same brightness to every pixel. Equal photon weighting of the 31 bands and unit quantum efficiency are assumed. No wavelength-dependent conversion from an absolute radiometric measurement is claimed.

## 2.2 Router and absorptive filter

The baseline expected detector counts are

$$\lambda^{\mathrm{R}} = N\eta P_K x, \tag{2}$$

$$\lambda^{\mathrm{F}} = \frac{N\tau}{K} P_K x. \tag{3}$$

Here $\eta = 0.84$ is router throughput and $\tau = 1$ is in-band filter transmission. Each filter occupies $1/K$ of the fixed macro-pixel area, which produces the extra factor in Eq. (3). With equal-width bands and a white input, the per-channel means are approximately $N\eta/K$ and $N\tau/K^2$, respectively.

The 84% value motivates a throughput scenario; it is not a measured efficiency for the full family of three- to 24-channel devices. Nor does routing classify a mixed beam by its perceived color. Different wavelength components contribute linearly to their respective outputs. The decoder later estimates their mixture.

Each output is assumed to observe the same local scene spectrum. Spatial aliasing, blur, demosaicing and angular response are outside the model. Consequently, the study compares spectral estimation at a fixed spatial sampling unit rather than complete camera image quality.

## 2.3 Photon detection and coding

In the baseline, independent counts satisfy $c_j \sim \mathrm{Poisson}(\lambda_j)$. Write $t = \eta$ for routing and $t = \tau/K$ for filtering, and let $w_j = (P_K\mathbf{1})_j$. The white-reference count is $Ntw_j$. Normalized measurements are $z_j = c_j/(Ntw_j)$.

The uniform ADC uses headroom $H = 1.2$ and $L = 2^B$ levels:

$$Q_B(z) = \frac{H}{L-1}\,\mathrm{round}[(L-1)\,\mathrm{clip}(z/H,0,1)]\,. \tag{4}$$

ADC gain is calibrated separately for each exposure, optical mode and channel count. It does not depend on the held-out scene. This is an exposure-dependent gain model, not one fixed full-well range spanning the entire sweep.

Two alternative code distributions are considered. The square-root ADC uniformly quantizes $\sqrt{\mathrm{clip}(z/H,0,1)}$ and squares the decoded value. The empirical quantile ADC places thresholds at training-distribution quantiles and uses the within-cell training mean as each reconstruction value. Nonuniform scalar quantization is established practice [8]; its role here is a controlled change in how the same output bits are used. Section 3.3 gives the calibration details.

## 2.4 Throughput and information

For fixed $K$, the router and filter have identical mean count vectors when the filter exposure is increased to

$$N_{\mathrm{F}} = \frac{\eta K}{\tau} N_{\mathrm{R}}. \tag{5}$$

With the same spectral operator, count statistics and ADC calibration, their measurement distributions are then identical. This follows directly from Eqs. (2)–(3) and was checked numerically using common random draws. Thus, the modeled routing advantage is an exposure-efficiency advantage, not an additional information gain at equal detected counts. The reconstruction error at equal incident exposure still requires evaluation because it is a nonlinear function of noise, coding and the spectral prior.

# 3 Reconstruction and selection

## 3.1 Decoders

The main decoder is an affine ridge model trained on simulated noisy, quantized measurements. Features are standardized using training means and standard deviations. With standardized feature matrix $Z$, the fitted weights solve

$$\widehat{W} = \arg\min_{W} \|ZW - X_c\|_F^2 + \alpha\|W\|_F^2, \tag{6}$$

where $X_c$ contains centered training spectra. The training mean is restored after prediction and values are clipped to $[0,1]$. Regularization is selected from $\alpha \in \{10^{-4}, 10^{-2}, 1, 100\}$ on validation scenes.

Three controls test reconstruction dependence. A clean-trained affine model uses noiseless, unquantized training measurements and $\alpha = 10^{-5}$. A quadratic

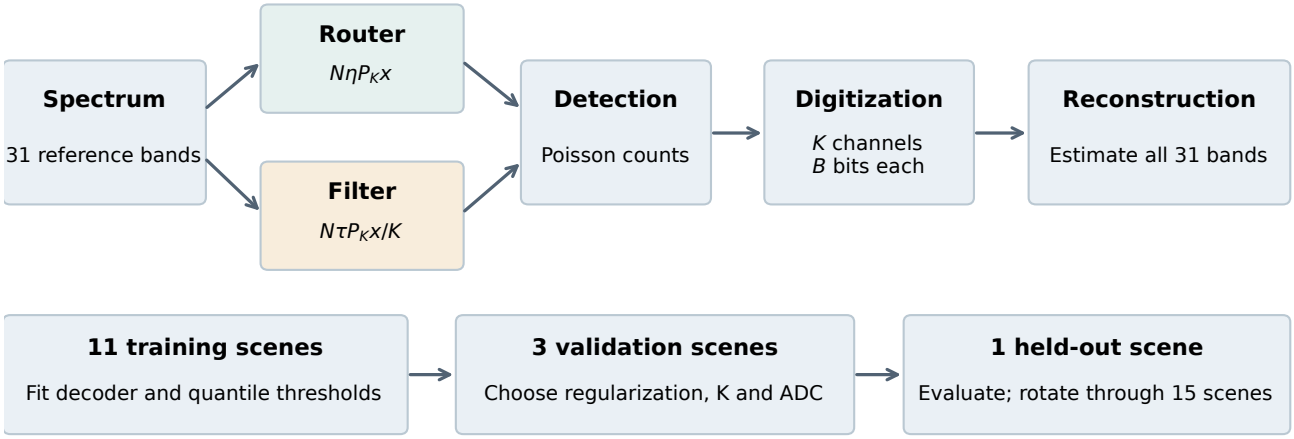



**Figure 1.** Simulation and evaluation protocol. Router and filter are alternative acquisition paths. Both produce $K$ noisy measurements followed by $B$-bit coding. Training fits the decoder and, where applicable, ADC thresholds. Validation selects the design. The held-out scene is used only to evaluate it. The macro-pixel area and white-reference exposure are fixed across allocations.

ridge model augments measurements with their squares and adjacent-channel products; it uses the same noisy training and regularization selection as the main model. Piecewise-linear interpolation between channel centers uses no learned spectral prior. A training-mean spectrum provides a measurement-free baseline. These controls are deliberately interpretable; they do not represent the best available neural reconstruction methods.

### 3.2 Scene-held-out protocol

The supplied CAVE subset contains 15 complete scenes with 31 bands from 400 to 700 nm [4]. Every spectral PNG was checked for completeness and consistent dimensions. Files are decoded as 16-bit values and divided by 65,535. No individual spectrum or scene is rescaled to unit brightness. The exact scene list is given in Appendix A.

Each scene is tested once. In alphabetical scene order, the next three cyclic scenes form validation data and the remaining eleven form training data. A fixed random sample without replacement gives 600 training pixels per training scene, 1,024 validation pixels per validation scene and 4,096 test pixels. Two independent noisy copies of the 6,600 training spectra are used to fit the matched decoders. Validation uses one noise realization; testing uses three. All candidate methods share sampled pixels and, where possible, the same underlying photon draws.

For a fixed exposure and optical mode, the selector first chooses the ridge parameter for each design and then chooses the design with the smallest mean validation-scene RMSE. The joint selector extends the design set to include uniform, square-root and quantile ADCs. The fixed-$K = 3$ calibrated control may also choose its ADC using validation. None of these decisions uses the held-out scene. A separate test-error minimum is reported only as a descriptive summary of the evaluated candidates.

### 3.3 Training-only quantile calibration

For channel $j$, let $\widehat{F}_j$ be the empirical distribution of its normalized noisy training counts. Candidate thresholds are

$$u_{j,\ell} = \widehat{F}_j^{-1}(\ell/L), \qquad \ell = 1, \ldots, L-1. \tag{7}$$

Duplicate thresholds are merged; the quantizer therefore uses at most $L$ codes. Values exactly on a threshold enter the lower cell. This matters for low-count Poisson data: when the median is zero, zero and positive counts must remain distinguishable. Each occupied cell is decoded to the mean of its training samples; empty cells use the midpoint of their finite training boundaries. The two outer cells extend to the tails of the input domain.

Thresholds and decoded values use training data only. They are shared calibration parameters, not additional transmitted bits for each pixel. Their storage and analog implementation costs are not included in the 24-bit output constraint. Quantile calibration aims to distribute code usage more evenly; it does not directly minimize spectral reconstruction error. In particular, marginal code entropy measures neither independence between channels nor information about the unknown spectrum.

### 3.4 Outcomes and uncertainty

For $n$ test pixels, spectral RMSE is

$$\mathrm{RMSE} = \sqrt{\frac{1}{31n}\sum_{i=1}^{n} \|x_i - \widehat{x}_i\|_2^2}. \tag{8}$$

The secondary metrics are mean spectral angle (SAM, degrees) and PSNR with peak value one. Zero reference vectors are excluded from SAM; a nonzero reference predicted as zero is assigned 90°. Very dark nonzero spectra remain included. Metrics are computed per scene and trial, averaged across trials, and

then averaged equally across scenes. PSNR is averaged after its per-scene calculation, not calculated from the mean RMSE.

Paired uncertainty intervals use 10,000 bootstrap resamples of the 15 scenes after trial averaging. For a reference $a$ and candidate $b$, relative gain is $100(\overline{R}_a - \overline{R}_b)/\overline{R}_a$; the ratio is recomputed within each resample. These intervals describe variation in this subset. Overlapping training sets, a nonrandom scene subset and exploratory comparisons prevent interpreting them as definitive population-level confidence intervals. No multiplicity correction is applied.

# 4 Experiments and results

The primary exposure grid contains 10, 30, 100, 300, 1,000, 3,000, 10,000, 30,000 and 100,000 white-reference photons. The initial robustness sweep contains 72,090 trial–condition–decoder records across 12 configurations. The subsequent quantile control adds 9,720 records using matched affine and quadratic decoders. Records sharing a scene are not independent experimental samples. The extension was motivated by inspecting the original calibration sensitivity and is reported as exploratory.

## 4.1 Uniform-ADC allocation

Figure 2 shows that the largest channel count is not automatically useful. Under the matched affine decoder, the router's lowest mean test RMSE occurs at $6 \times 4$ for most sampled exposures, at $8 \times 3$ for $N = 30$, and at $4 \times 6$ for $N = 3{,}000$. The filter favors $3 \times 8$ through $N = 3{,}000$ and $4 \times 6$ at the three largest exposures. Several neighboring designs are close; the cell with the smallest number should not be mistaken for an accurately located physical transition.

The decoder control changes this picture (Fig. 3). A model trained only on clean measurements produces a different allocation map from one trained on the noise and quantization present at evaluation. The quadratic model changes the preference again. This is expected when a learned estimator can exchange variance for bias, but it matters for the interpretation of a budget sweep: the selected $K$ belongs to the complete acquisition–reconstruction system.

## 4.2 Performance on held-out scenes

The validation-selected comparison is the practical test of the design procedure. Figure 4 and Table 1 report its performance on unseen scenes. At $N = 10^5$, selecting $K$ under uniform coding reduces router RMSE from 0.0294 to 0.0257. Allowing the selector to choose the ADC as well reduces it further to 0.0223. The total reduction relative to fixed $3 \times 8$ is 24.0%, with a paired bootstrap interval of 13.9–33.1%. The incremental reduction relative to the uniform-ADC selector is 13.1% (8.4–17.8%).

This comparison should not give the three-channel baseline an avoidable calibration disadvantage. When that baseline can also select its ADC, the joint router selector retains a 24.2% reduction at $N = 10^5$. Thus, the improvement in this setting is not explained solely by replacing the three-channel ADC. At low exposure the gains are smaller, and some validation choices transfer poorly. Figure 5 retains these cases rather than reporting only conditions with a positive effect.

The filter follows the same broad pattern at high exposure. Joint selection reaches RMSE 0.0230, a 21.8% reduction from its fixed $3 \times 8$ uniform baseline. For both optical modes, the high-exposure joint result comes from the square-root option rather than the added quantile ADC. The latter therefore provides a useful control, not an additional performance claim.

## 4.3 Code use and reconstruction error

For a one-bit uniform ADC with $H = 1.2$, the decision threshold is 0.6 in normalized channel intensity. Many scene channels lie well below this level. As exposure grows, shot noise shrinks around their mean, and repeated measurements can settle into the same code without resolving differences within that interval. More incident light therefore need not produce a monotonic reduction in reconstruction error at a fixed coarse threshold.

The quantile control directly tests this explanation. At $N = 10^5$, its mean marginal code entropy for the $24 \times 1$ router is 0.834 bits per channel, compared with 0.144 for the uniform ADC. The corresponding mean RMSE changes from 0.1209 to 0.1168. This shows that part of the one-bit failure is a threshold-placement effect.

However, balancing code occupancy is not sufficient to choose the sensor. Quantile cells devote many levels to the dense part of the training intensity distribution and fewer to its tail. Distribution shifts between scenes and interactions with the decoder can make that allocation unfavorable for squared spectral error. At high exposure the quantile-only selector has RMSE 0.0316, worse than 0.0257 for the uniform selector. This result is consistent with its calibration objective: equalizing marginal occupancy does not optimize the final reconstruction loss.

## 4.4 Routing and sensor sensitivity

With a uniform ADC and validation-selected channel count, the router reduces mean error relative to the filter by 24.6% at $N = 100$ and 1.7% at $N = 10^5$. These values refer to equal incident exposure and the stated area model. Their direction is partly built into the throughput assumptions. Equation (5) makes that limitation explicit. At high exposure, representation and coding errors account for more of the remaining loss, so recovering additional photons offers less improvement.

The original sweep varied ADC headroom, read noise, dark signal, router crosstalk, throughput, spectral binning and total output budget. Appendix B gives

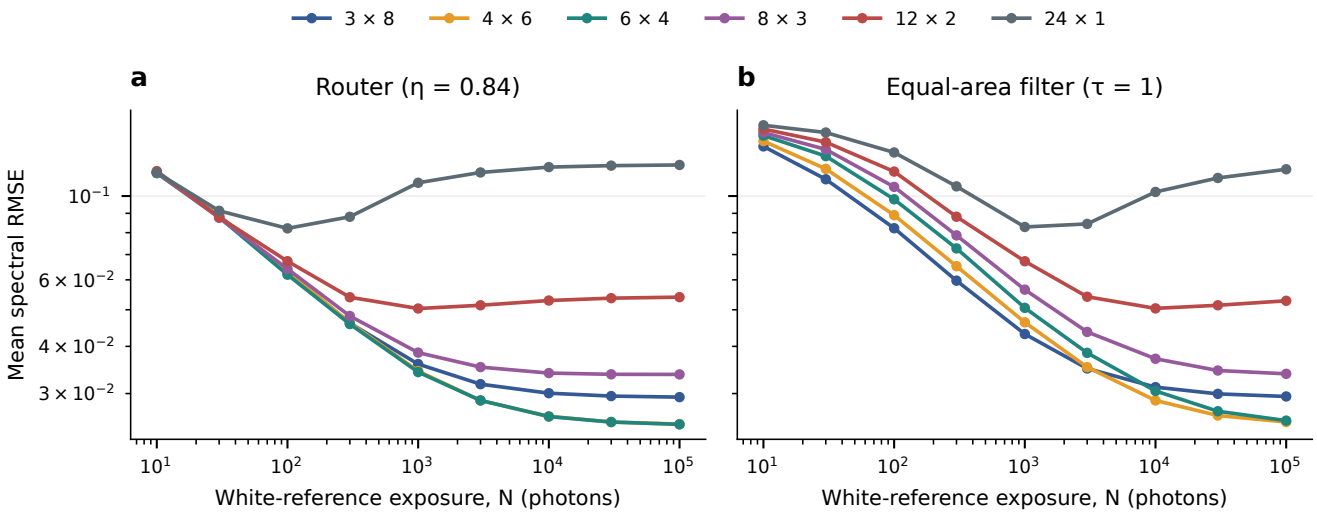



**Figure 2.** Mean held-out spectral RMSE under uniform coding and the matched affine decoder. Each curve represents one 24-bit allocation. More channels can help, but $24 \times 1$ performs poorly with these white-reference thresholds. The curves are conditional on this ADC and decoder; minima across test curves are descriptive, not a deployable selection procedure.

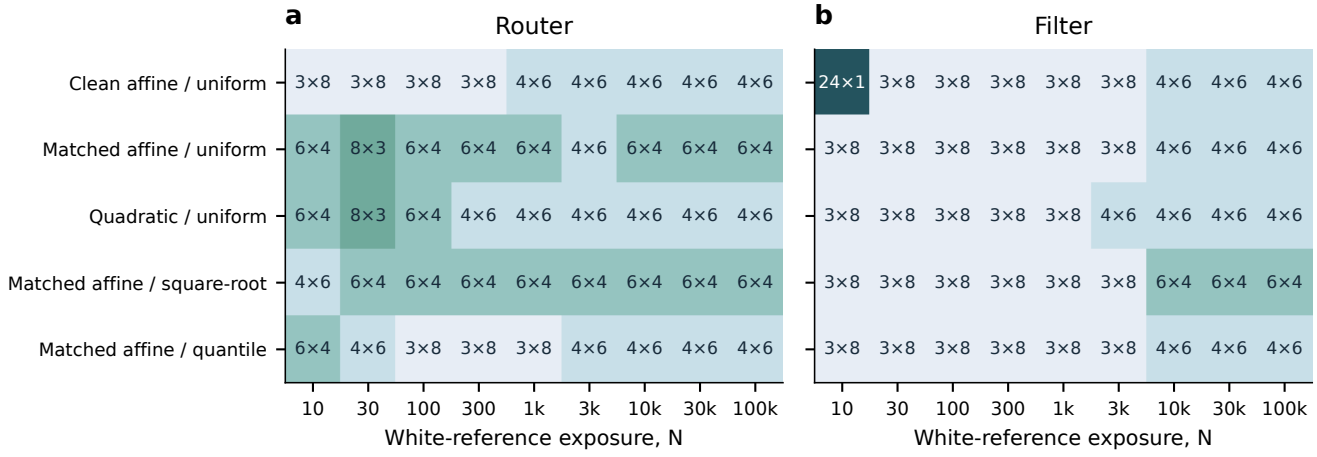



**Figure 3.** Descriptive allocations with the lowest mean test RMSE. Rows change the decoder or quantizer while holding the 24-bit output constraint fixed. The upper three rows isolate reconstruction under uniform coding; the lower rows change the code distribution with the matched affine decoder. Each cell is $K \times B$. These maps summarize the test results and are not used to choose a design for the held-out scene.

the conditions and allocation maps. These tests reinforce the conditional nature of the recommendation. They are individual perturbations and one combined stress case, not a full factorial characterization of a physical sensor. The new quantile ADC has not been rerun under every sensor imperfection.

## 4.5 Spectral rank and a compact rule

The original hypothesis was that a scene's intrinsic spectral dimension would predict its useful channel count. Effective rank was estimated from the covariance eigenvalues of the reference spectra. In the revised matched-affine experiment, its association with the scene's high-exposure test-optimal $K$ is weakly resolved: Spearman $\rho = 0.355$, with unadjusted permutation $p = 0.198$. Normalizing each spectrum by its sum changes the association. This dataset and protocol therefore do not support a reliable rank-to-channel rule.

A more modest rule fits validation error directly. For each allocation, a nonnegative model

$$\widehat{R^2}_K(N) = C_K + \frac{A_K}{N} \tag{9}$$

is fitted at $N = 30$, 1,000 and 30,000. The preferred $K$ minimizes this predicted error. At the six remaining exposures, the rule reduces the mean RMSE pooled equally over scenes and exposures by 2.48% for routing and 1.33% for filtering relative to fixed $3 \times 8$. It trails the full validation sweep and is only an approximate calibration shortcut. Decoders are still fitted at every evaluated exposure; this experiment does not establish a complete three-exposure training procedure. Appendix C explains the approximation and its failure modes.

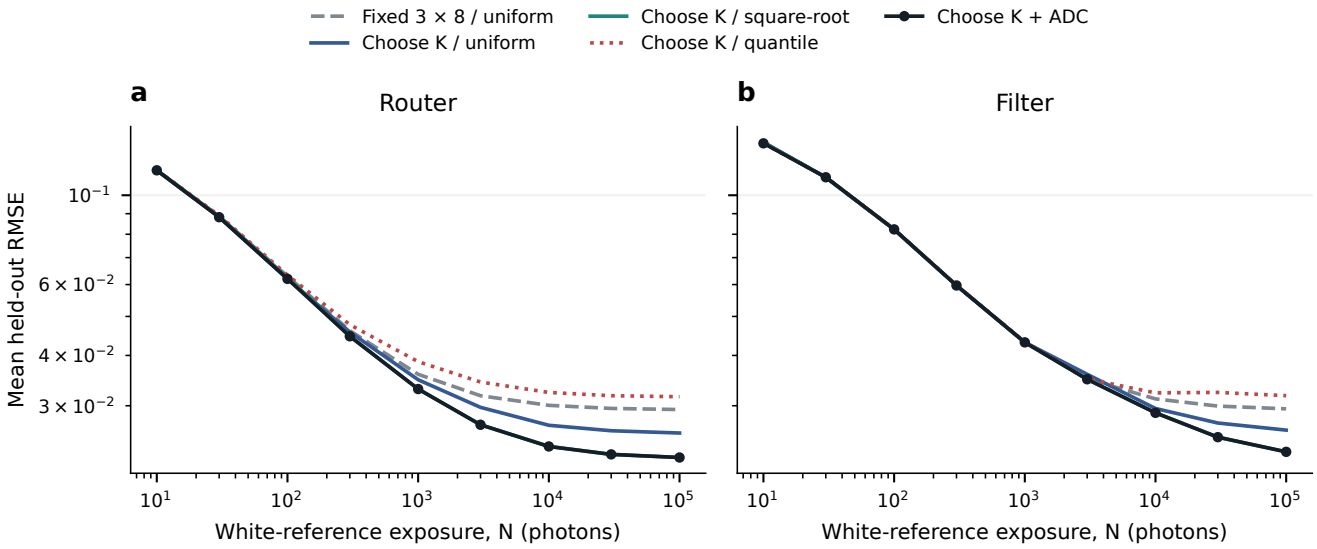



**Figure 4.** Performance of designs chosen using validation scenes. All curves use the matched affine decoder. Joint selection chooses both $K$ and ADC type; each single-ADC selector chooses only $K$. Overlapping curves are retained: at high exposure, the joint selector chooses the square-root ADC in every fold. The quantile control is included even where it performs worse.

**Table 1.** Held-out results after validation-only selection. RMSE is in normalized spectral units; SAM is in degrees and PSNR in dB. Entries average three noise trials within each scene and then weight the 15 scenes equally. The joint selector chooses channel count and ADC type.

| | | Router | | | Filter | | |
|---|---|---|---|---|---|---|---|
| $N$ | Selection procedure | RMSE ↓ | SAM ↓ | PSNR ↑ | RMSE ↓ | SAM ↓ | PSNR ↑ |
| 100 | Fixed 3 × 8, uniform | 0.0629 | 24.10 | 24.26 | 0.0822 | 25.44 | 21.94 |
| | Choose $K$, uniform | 0.0620 | 24.35 | 24.39 | 0.0822 | 25.44 | 21.94 |
| | Choose $K$ and ADC | 0.0620 | 24.25 | 24.39 | 0.0823 | 25.44 | 21.94 |
| 1,000 | Fixed 3 × 8, uniform | 0.0359 | 18.11 | 29.22 | 0.0431 | 20.07 | 27.55 |
| | Choose $K$, uniform | 0.0348 | 18.58 | 29.42 | 0.0431 | 20.07 | 27.55 |
| | Choose $K$ and ADC | 0.0330 | 17.15 | 29.87 | 0.0431 | 20.14 | 27.56 |
| 10,000 | Fixed 3 × 8, uniform | 0.0301 | 16.25 | 30.95 | 0.0312 | 16.67 | 30.58 |
| | Choose $K$, uniform | 0.0268 | 16.72 | 31.81 | 0.0295 | 17.14 | 30.94 |
| | Choose $K$ and ADC | 0.0238 | 13.78 | 32.81 | 0.0288 | 15.48 | 31.12 |
| 100,000 | Fixed 3 × 8, uniform | 0.0294 | 15.94 | 31.20 | 0.0295 | 16.00 | 31.16 |
| | Choose $K$, uniform | 0.0257 | 16.30 | 32.22 | 0.0261 | 16.46 | 32.07 |
| | Choose $K$ and ADC | 0.0223 | 13.19 | 33.40 | 0.0230 | 13.57 | 33.09 |

# 5 Discussion

The practical result is a change in how the original question should be asked. "How many channels at 24 bits?" is incomplete unless the ADC thresholds, expected exposure and reconstruction model are specified. A useful design procedure can remain simple: define physically plausible candidates, calibrate each without the evaluation scenes, choose using validation error, and report the chosen design's performance on a separate scene. The contribution lies in the controlled comparison and its measured consequences, not in the use of a finite search itself.

The quantizer extension also narrows the interpretation of the one-bit result. Poor performance with white-reference thresholds does not establish that one-bit spectral sensing is intrinsically ineffective. Replacing those thresholds changes the code distribution markedly, but improves high-exposure one-bit RMSE only modestly. More balanced code use does not make the quantile ADC the best reconstruction option. Channel count, intensity distribution and the learned spectral prior have to be assessed together.

Several limitations affect transfer to hardware. The router and filter share idealized contiguous spectral responses; measured wavelength leakage, angular response and fabrication variation are absent. Constant router efficiency is especially optimistic for an arbitrary number of channels. The 24-bit constraint omits circuit area, power, calibration storage and readout bandwidth overhead. Exposure-dependent gain assumes that an appropriate operating range is known before acquisition. A fixed manufactured router cannot change its spectral channel layout after capture; the selector is a design or calibration tool unless reconfigurable optics are added.

The dataset provides another limitation. Fifteen indoor scenes are sufficient to expose dependence on

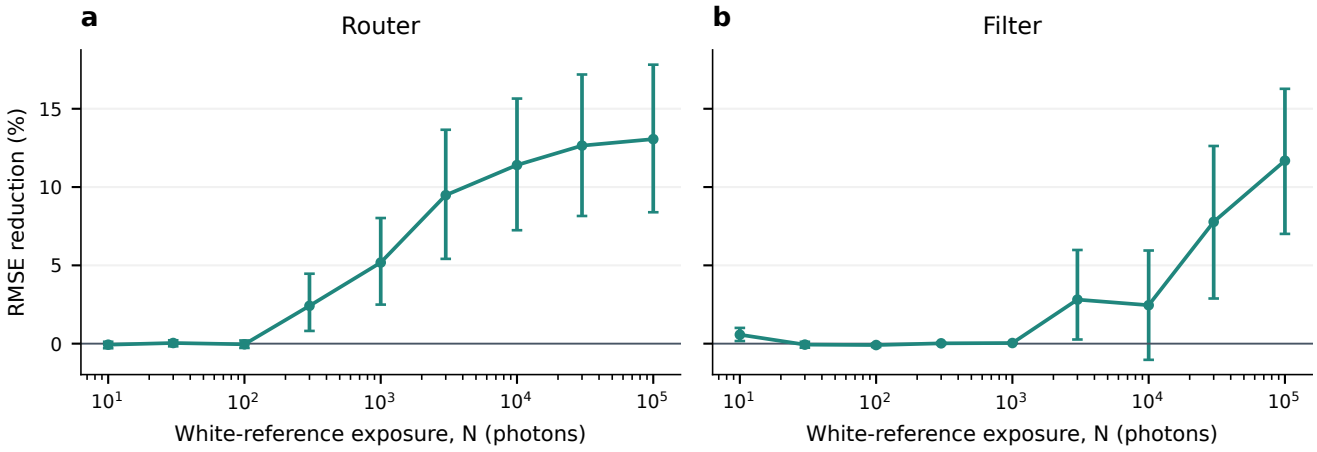



**Figure 5.** Incremental value of allowing ADC choice after channel count can already be selected. Positive values indicate lower mean RMSE for joint selection than for the uniform-ADC selector. Bars are paired 95% scene-bootstrap intervals. The uncertainty unit is the scene, not the individual spectrum or noise trial.

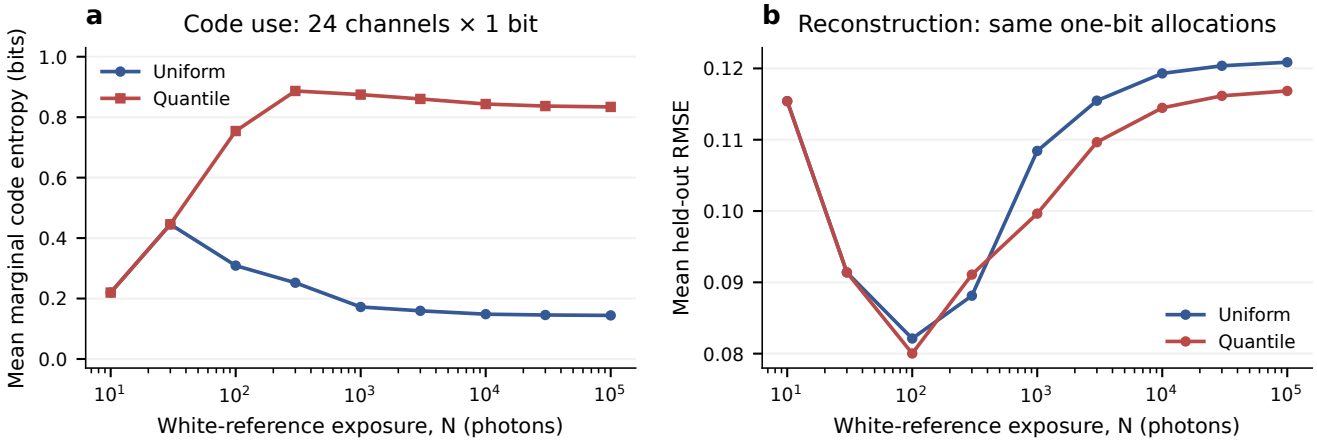



**Figure 6.** One-bit diagnostic for the router. Left: average per-channel code entropy on held-out scenes. Right: reconstruction RMSE for the same $24 \times 1$ design. Quantile thresholds are fitted on training data only. More balanced code use reduces high-exposure RMSE only modestly and does not make this the best 24-bit design. The full quantile selector also fails to outperform the square-root selector. Entropy here is a marginal occupancy statistic, not mutual information.

the modeling choices, but they do not establish performance on outdoor scenes, unfamiliar illuminants or material spectra from another instrument. The training and validation scenes are reused across outer folds. Both the original robustness sweep and the quantile extension followed inspection of earlier results. Their uncertainty estimates are therefore descriptive, and an independent dataset is needed before turning the resulting allocations into a general recommendation.

Finally, the 84% routing scenario should not be presented as an experimentally demonstrated camera improvement. The optical literature establishes that wavelength routing is possible [6, 7]; this work estimates downstream consequences under a simplified model. A physical follow-up would replace $P_K$, throughput and detector-noise assumptions with measured quantities while retaining the same separation between calibration, selection and evaluation.

# 6 Conclusion

At a fixed 24-bit output budget, the most useful number of spectral channels depends on photon throughput, ADC calibration and reconstruction. A 15-scene held-out study shows that selecting the quantizer together with channel count can improve reconstruction beyond a channel-only choice. The added quantile control demonstrates that a poor one-bit result can partly arise from threshold placement, while also showing that code occupancy alone is an unreliable optimization target. No stable intrinsic-rank rule is established. The supported design rule is to calibrate and select the complete measurement pipeline against held-out spectral error, under the exposure and hardware constraints that the intended sensor will actually face.

## Data and code availability

The accompanying source package contains the experiment scripts, sampled spectra, scene folds, calibration parameters, raw trial metrics, analysis tables and figure-generation code. Input checksums identify the 15-scene CAVE archive; the original full-resolution images are not redistributed in this package. The source spectra are from the CAVE dataset [4].

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# A Reproducibility and evaluation details

## A.1 Scenes, folds and randomization

The 15 scenes are balloons, beads, cd, chart and stuffed toy, clay, cloth, Egyptian statue, feathers, flowers, glass tiles, pompoms, sponges, stuffed toys, superballs and thread spools. Exact folder names are recorded in the supplied fold definitions. The source archive contains 465 spectral PNGs. This is the complete set of usable scenes in that archive, not the complete CAVE database.

The outer folds use the alphabetical list above. For test scene $i$, validation scenes have cyclic indices $i+1$, $i+2$ and $i+3$; all remaining scenes train the model. The same 4,096 sampled spectra per scene are cached. Training and validation use the first 600 and 1,024 cached spectra, respectively, when a scene takes those roles. The roles are disjoint within every fold. The sample and noise seeds are derived by SHA256 from the fixed base seed 20261013 and explicit identifiers for scene, optical mode, exposure, channel count and draw. Noise streams are shared across quantizers so that calibration comparisons do not depend on different photon realizations.

The initial robustness analysis uses bootstrap seed 20261013; the extension uses seed 20261014. The latter is a separate resampling seed, not an additional source dataset or an additional acquisition. Complete fold memberships and exact sampled spectra are provided in the accompanying files.

## A.2 Metrics and diagnostic checks

The spectral angle for a nonzero reference and nonzero estimate is $\arccos[(x^\top \hat{x})/(\|x\|_2\|\hat{x}\|_2)]$, converted to degrees. The cosine argument is clipped to $[-1,1]$ for numerical stability. PSNR equals $-20\log_{10}(\mathrm{RMSE})$ for each scene/trial because the peak value is one. The main outcome remains RMSE; channel selection is not separately optimized for SAM or PSNR.

Ten checks accompany the original simulator, including photon conservation, crosstalk conservation, output-bit accounting, Poisson sampling, ADC code limits and the zero-prediction convention for SAM. Four extension checks verify monotonic quantization with no more than $2^B$ codes, preservation of zero-versus-positive counts when the training median is zero, constant-input handling, and the detected-photon equivalence in Eq. (5). Duplicate result keys, missing conditions, nonfinite metrics and overlap between the training/validation/test scene sets are checked during analysis. These checks test the implementation; they do not validate the physical assumptions.

## A.3 Selection results at every exposure

**Table 2.** Incremental gain from joint channel/ADC selection over channel selection with a uniform ADC. Positive values favor joint selection. Paired scene-bootstrap intervals are descriptive.

| | Router | | | Filter | | |
|---|---|---|---|---|---|---|
| $N$ | Gain (%) | 95% interval | Scenes improved | Gain (%) | 95% interval | Scenes improved |
| 10 | -0.07 | [-0.29, 0.12] | 2/15 | 0.57 | [0.17, 1.00] | 12/15 |
| 30 | 0.04 | [-0.17, 0.21] | 5/15 | -0.06 | [-0.26, 0.12] | 9/15 |
| 100 | -0.04 | [-0.27, 0.20] | 3/15 | -0.09 | [-0.21, 0.00] | 2/15 |
| 300 | 2.41 | [0.81, 4.47] | 10/15 | 0.02 | [-0.03, 0.07] | 9/15 |
| 1,000 | 5.18 | [2.49, 8.02] | 13/15 | 0.04 | [0.00, 0.08] | 11/15 |
| 3,000 | 9.48 | [5.41, 13.66] | 14/15 | 2.81 | [0.26, 5.98] | 14/15 |
| 10,000 | 11.41 | [7.24, 15.65] | 14/15 | 2.46 | [-1.03, 5.95] | 12/15 |
| 30,000 | 12.65 | [8.15, 17.19] | 14/15 | 7.78 | [2.88, 12.62] | 12/15 |
| 100,000 | 13.06 | [8.39, 17.82] | 14/15 | 11.68 | [7.01, 16.27] | 13/15 |

The scene counts in Table 2 record strictly lower mean test RMSE; ties are not counted as improvements. Joint selection searches 18 acquisition candidates (six allocations and three ADC types) for the matched affine decoder. The larger candidate set can overfit the three validation scenes. Its test performance, including negative changes, is retained. No per-test-scene switching of methods is allowed after inspecting the test error.

# B Sensor sensitivity and supplementary controls

Read noise is added after Poisson detection. For a mean dark signal $d$, counts are sampled from Poisson$(\lambda + d)$ and the expected dark offset $d$ is subtracted. Negative values are clipped to zero before coding. This is a simplified offset-correction model. Crosstalk uses a column-stochastic adjacent-channel matrix $M$ in the router

**Table 3.** Conditions in the original 12-configuration sweep. Unlisted quantities retain their baseline values. The quantile extension is a separate baseline-physics experiment.

| Condition | Setting and interpretation |
|---|---|
| Baseline | 24 bits; $H = 1.2$; $\eta = 0.84$; $\tau = 1$; no read noise, dark signal or crosstalk. |
| Headroom 1.0 / 2.0 | Two separate configurations changing only the uniform ADC range. |
| Square-root ADC | Fixed nonlinear code distribution with the same number of output codes. |
| Read noise | Independent Gaussian noise with standard deviation 2 electrons per output. |
| Combined stress | Read noise 5 electrons; mean dark signal 0.2 electrons; 15% router crosstalk. |
| Crosstalk | 15% of a channel's light redistributed to adjacent outputs, conserving total counts. |
| Declining throughput | Hypothetical router efficiency $\eta(K) = 0.84 - 0.24(K-3)/21$. |
| Filter transmission | In-band transmission $\tau = 0.8$. |
| Equal-width bands | Fractional overlap between source bins and $K$ equal-width output bands. |
| 12-bit budget | Allocations $(3,4)$, $(4,3)$, $(6,2)$ and $(12,1)$. |
| 48-bit budget | Allocations $(3,16)$, $(4,12)$, $(6,8)$, $(8,6)$, $(12,4)$, $(16,3)$ and $(24,2)$. |

mean $N\eta MP_K x$; the decoder is calibrated to the same known mixing. None of the stress parameters is presented as a measured property of a fabricated device.

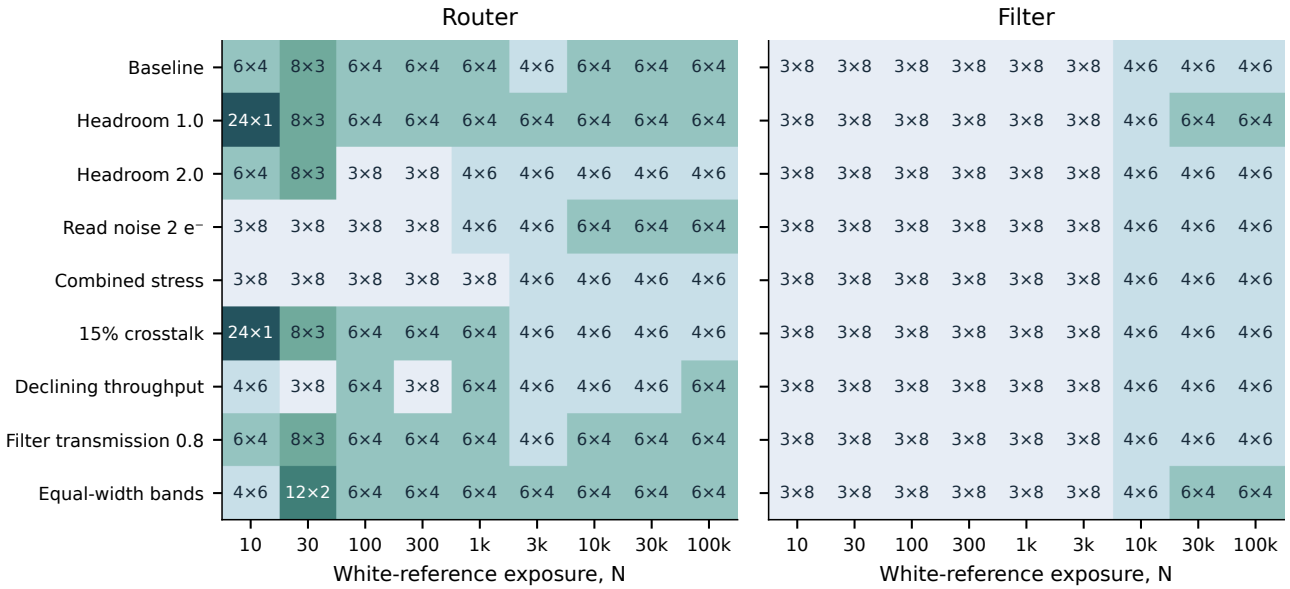



**Figure 7.** Sensitivity of the descriptive test-error minimum under the matched affine decoder and uniform ADC. The full numerical records include RMSE, SAM and PSNR, rather than only the winning allocation. Near-ties and changes between sampled exposures should not be interpreted as sharp physical boundaries.

# C Approximate risk and spectral complexity

The inverse-exposure term in Eq. (9) follows from the baseline count normalization. If $\mu_j = (P_K x)_j / w_j$, then

$$\mathrm{Var}(z_j \mid x) = \frac{\mu_j}{N t w_j}. \tag{10}$$

For a fixed affine decoder with effective unstandardized rows $W_j$, the corresponding mean spectral error contribution is

$$\frac{1}{31N} \sum_j \frac{\|W_j\|_2^2 \, \mathbb{E}[\mu_j]}{t w_j}. \tag{11}$$

This suggests an inverse-$N$ component above a representation and quantization floor. At sufficiently high bit depth, a uniform quantizer with step $\Delta = H/(2^B - 1)$ is sometimes approximated by uncorrelated noise of variance $\Delta^2/12$ [8]. That approximation is not valid as a general description of one-bit measurements. Clipping, signal-dependent quantization error and exposure-dependent decoder weights also break the simplest model. Accordingly, $C_K + A_K/N$ is fitted empirically and tested, rather than treated as a physical theorem.

Effective rank is $\exp[-\sum_i p_i \log p_i]$, where $p_i$ are the normalized eigenvalues of the centered spectral covariance. The raw statistic includes overall brightness variation. A separate shape statistic normalizes each spectrum by its

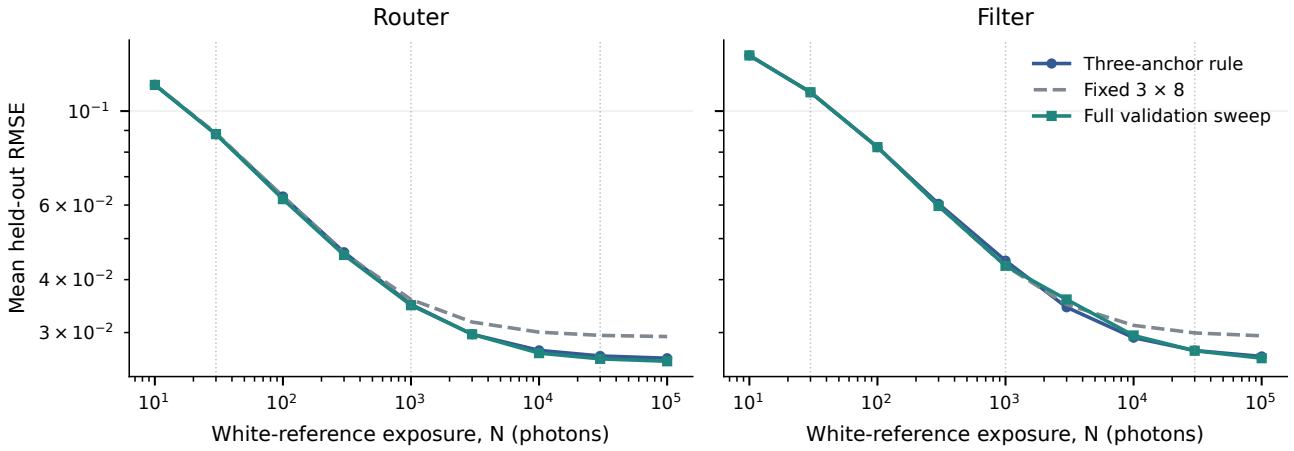



**Figure 8.** Three-anchor approximation under uniform coding and the matched affine decoder. Dotted vertical lines mark the exposures used to fit validation error. All other exposures assess interpolation or extrapolation. The coefficients are fitted separately within each scene fold and optical mode. This reduces the allocation-error calibration grid; the decoder training grid is unchanged.

spectral sum after excluding pixels with mean intensity at most $10^{-3}$. Associations use the test-scene rank and that scene's test-error-minimizing high-exposure channel count. These are exploratory descriptions, not variables supplied to the selector. Two-sided permutation tests use 20,000 permutations and a $+1$ correction. The lack of stable support for a rank rule is retained in the paper rather than replacing the hypothesis after seeing the outcome.

# D Visual reconstruction example

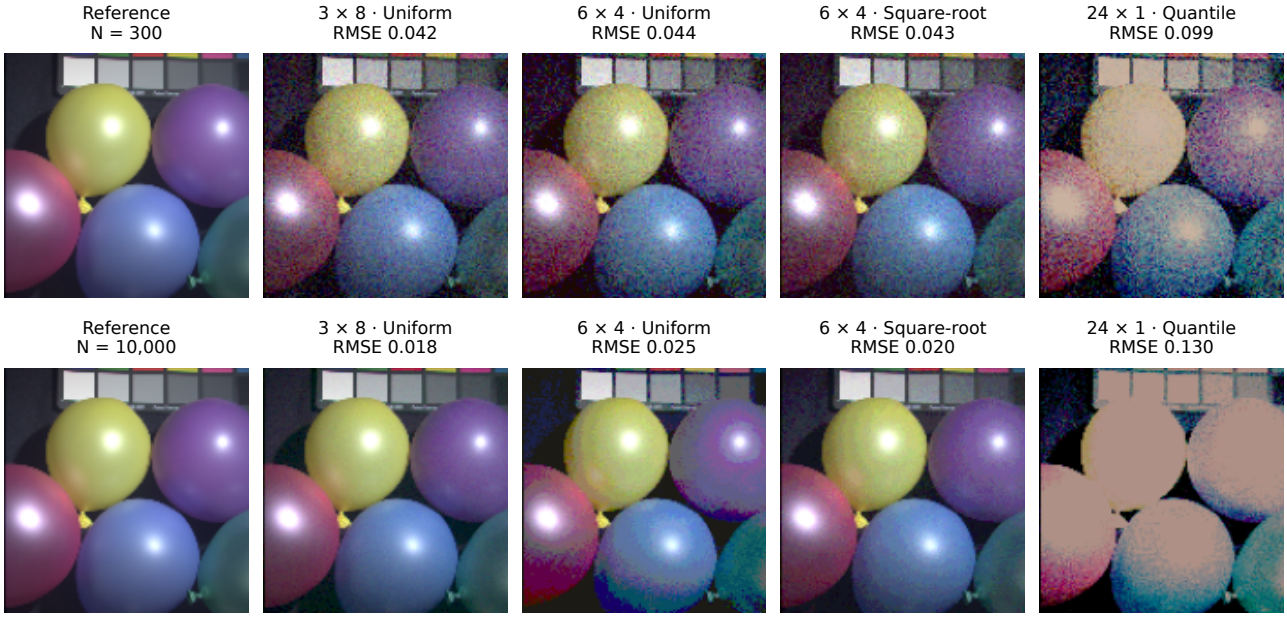



**Figure 9.** The held-out `balloons` scene at two exposures, using the matched affine decoder and router model. Every panel uses the same display transform: 650/550/450 nm bands, fixed gain 1/0.7 and gamma 1/2.2. This is a pseudo-color rendering, not a calibrated RGB photograph. Candidate panels are fixed comparisons, not selections made using the displayed scene's test error. Numbers above panels are RMSE over all 31 reconstructed bands in the displayed subsample.

The visual example uses every fourth source pixel in each spatial direction, giving a $128 \times 128$ display cube. The test scene is excluded from calibration and decoder fitting. Display pixels are not additional independent scene replicates, and the image results are not pooled into the quantitative cross-scene tables. Saved numerical cubes permit inspection beyond the three displayed wavelength bands.